

Hypothesis Testing (Statistics, Rejection regions, P-values)

Math 742

2026-03-02

Introduction

Last time, we viewed hypothesis testing as a decision problem $\rightarrow$ choosing rules that control risk.

Today we ask: What do those decision rules actually look like?

A hypothesis test must ultimately be a function of the observed data whose behavior we understand when the null hypothesis is true.

Main Idea: Hypothesis tests are functions of data, whose distributions we understand under H_0 .

Example:

Suppose your roommate wants to challenge you to a gambling game using their 1960 silver dollar. The gambling game will involve flipping this coin. You believe that your roommate is trying to swindle you out of money by using a coin that is weighted. To decide this you perform the following hypothesis test:

$$H_0 : p = 0.5, \text{ and } H_1 : p > 0.5$$

To perform the test, we flip the coin 15 times. The **rejection region** is the portion of the sample space where we no longer “believe” the null hypothesis. To find the rejection region, we assume the null hypothesis is true and then try and get a sense of how many heads would be “unreasonable”.

i.e. Reject if $T(X)$ lies in a specified region of the sample space.

```
pbinom(8,size=15,prob=0.5,lower.tail=FALSE) # Pr(more than 8 heads)
```

```
## [1] 0.3036194
```

```
pbinom(9,size=15,prob=0.5,lower.tail=FALSE) # Pr(more than 9 heads)
```

```
## [1] 0.1508789
```

```
pbinom(10,size=15,prob=0.5,lower.tail=FALSE) # Pr(more than 10 heads)
```

```
## [1] 0.05923462
```

```
pbinom(11,size=15,prob=0.5,lower.tail=FALSE) # Pr(more than 11 heads)
```

```
## [1] 0.01757813
```

Now we decide what is “unreasonable”. Based on the output, we could choose as our rejection rule, either one of these options:

$\rightarrow$ Reject if number of heads exceeds 10 (5.9% chance of seeing this with a fair coin)

$\rightarrow$ Reject if number of heads exceeds 11 (1.8% chance of seeing this with a fair coin)

Suppose we decide that anything that occurs 5.9% of the time or less, would make us reject the null hypothesis. Thus, our decision would be to reject H_0 if in 15 flips the number of heads exceeds 10.

This example points out our main idea from above. We have looked at a function of the data, whose distributions we understand under H_0 . In essence, we understand that if the coin were not biased in anyway, seeing more than 10 heads is not very likely.

Distributions as Decision Rules

From the previous example, we could write our rejection rule as:

Reject H_0 if $T(X) \in \{11,12,13,14,15\} \leftarrow$ this is our rejection region of the parameter space.

$T(X)$ is our test statistic where $T(X)$ equals the number of heads in 15 flips (i.e. the sum of 15 independent Bernoulli random variables).

What $T(X)$ does is reduce the high-dimensional data X to a single test statistic T . In summary, $T(X)$

- Summarizes the evidence against H_0 .
- Should remain unchanged or when transformed changes in a predictable way.
- In order to perform the hypothesis test, we need the distribution of $T(X)$ under the assumption that H_0 is true.

One sample Z-test

Imagine you're a data scientist at a tech firm optimizing a distributed inference system for a large language model (LLM). The system is claimed to have a population mean inference time of $\mu_0 = 200$ milliseconds (ms) per query, based on extensive historical benchmarks, with a known population standard deviation $\sigma = 30$ ms (derived from millions of prior runs under stable conditions). You collect a random sample of $n = 50$ inference times from a new deployment to test if the mean has increased (indicating potential degradation, e.g., due to network latency or model drift).

Suppose we want to test if the inference time exceeds 200.

- $H_0 : \mu \leq 200$
- $H_1 : \mu > 200$

Sample data (in ms, simulated for this example but representative of real telemetry):

[205, 198, 210, 195, 215, 202, 208, 190, 212, 200, 218, 197, 204, 209, 192, 214, 199, 206, 211, 193, 217, 196, 203, 207, 191, 213, 198, 205, 210, 194, 216, 201, 208, 189, 215, 197, 204, 209, 193, 211, 200, 206, 212, 195, 218, 199, 203, 207, 192, 214]

Test Statistic

Because we know the population standard deviation and we have a large sample size, the CLT tells us that the sample mean converges in distribution to: $N(\mu, \sigma^2/n)$. Or equivalently,

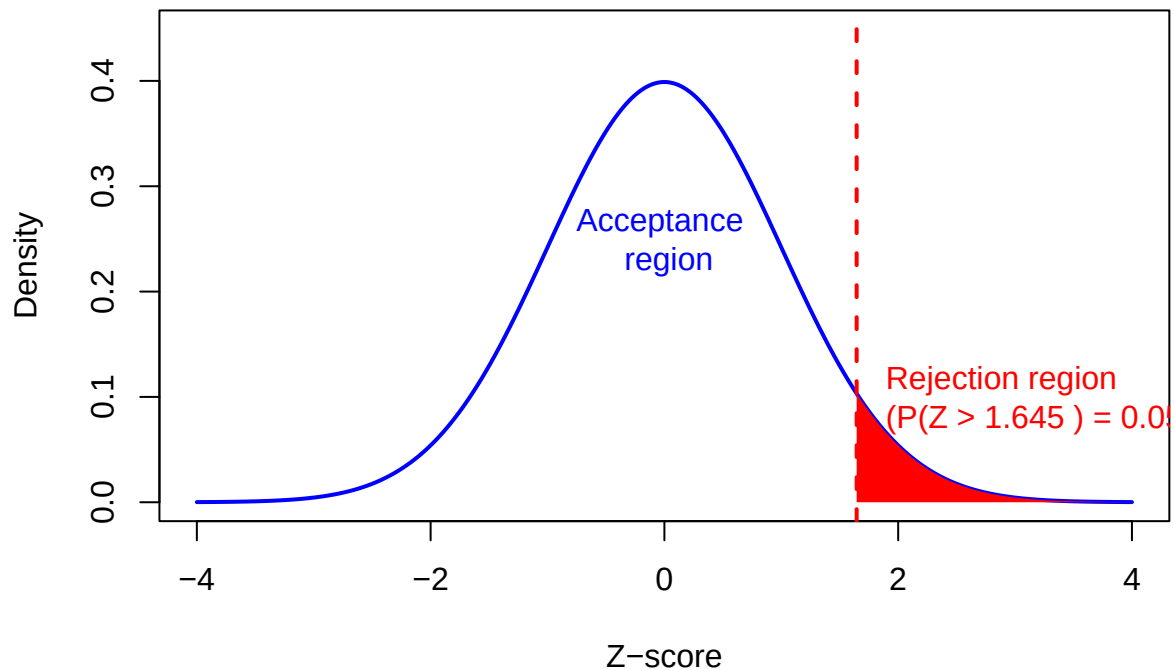
$$Z = \frac{\bar{x} - 200}{30/\sqrt{n}} \sim N(0, 1)$$

To be clear, we construct our test statistic Z in such a way that we know its distribution. Here, under H_0 , Z has a standard normal distribution (CLT).

Rejection Region

On a standard normal distribution the 5% upper tail cutoff would be 1.645

Standard Normal Distribution with Upper 5% Rejection Region



This means that if we get a standardized mean larger than 1.645 we would reject H_0 and conclude that the mean time could be no more than 200.

```
time <- c(205, 198, 210, 195, 215, 202, 208, 190, 212, 200, 218, 197, 204, 209,
192, 214, 199, 206, 211, 193, 217, 196, 203, 207, 191, 213, 198, 205, 210, 194,
216, 201, 208, 189, 215, 197, 204, 209, 193, 211, 200, 206, 212, 195, 218, 199,
203, 207, 192, 214)
Z <- (mean(time)-200)/(30/sqrt(50))
Z
```

```
## [1] 0.9475231
```

Conclusion: Our test statistic 0.9475 which is in the acceptance region, so we do not reject H_0 . We do not have sufficient evidence that the mean exceeds 200ms.

Two sample t-test

In data science, the independent two-sample t-test is widely used to compare the means of two groups when assessing differences in model performance, A/B testing outcomes, or treatment effects in observational data. It is appropriate under these assumptions:

- Observations are independent within and between groups.
- Data in each group are approximately normally distributed (check via Q-Q plots or Shapiro-Wilk; robust for larger samples via CLT).
- Variances may be equal (pooled t-test) or unequal (Welch's t-test, more robust and recommended by default).

The test statistic follows a t-distribution with degrees of freedom (df) approximated by the Welch-Satterthwaite equation if variances differ.

Scenario: A/B Testing Inference Latency in Machine Learning Models.

You are optimizing inference speed for a computer vision model deployed in a real-time edge application (e.g., autonomous drones). Two versions are tested:

- Group A (Control): Standard model, expected mean inference time $\sim 200\text{ms}$.
- Group B (Treatment): Optimized model (e.g., with quantization and pruning), hypothesized to be faster.

You collect independent random samples of inference times (in ms) from separate production runs:

Group A: $n_A = 40$ Group B: $n_B = 45$

Hypotheses (two-tailed, as we expect optimization to reduce latency):

$$H_0 : \mu_A - \mu_B = 0 \text{ (i.e. no difference in mean times)}$$

$$H_1 : \mu_A - \mu_B \neq 0 \text{ (something has changed, i.e. it's different)}$$

The samples yielded the following summary statistics.

Sample statistics:

- Group A mean: $\bar{x}_A = 193.44\text{ms}$, Group A sample sd: $s = 28.58\text{ms}$
- Group B mean: $\bar{x}_B = 184.32\text{ms}$, Group B sample sd: $s = 33.20\text{ms}$

Test Statistic

For this test, we only know the sample standard deviations so according to William Gosset, the difference in means follows a t-distribution as long as the populations are Normally distributed.

A basic rule of thumb is that if the standard deviations aren't off by more than a factor of 2 we can do a "pooled" t-test. This means that we can standardized our test statistic by using a pooled estimate of the standard deviation (s_p).

$$s_p = \sqrt{\frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}} = \sqrt{\frac{(40 - 1)(28.58)^2 + (45 - 1)(33.20)^2}{40 + 45 - 2}} = 31.11$$

Our standardized test statistic is thus,

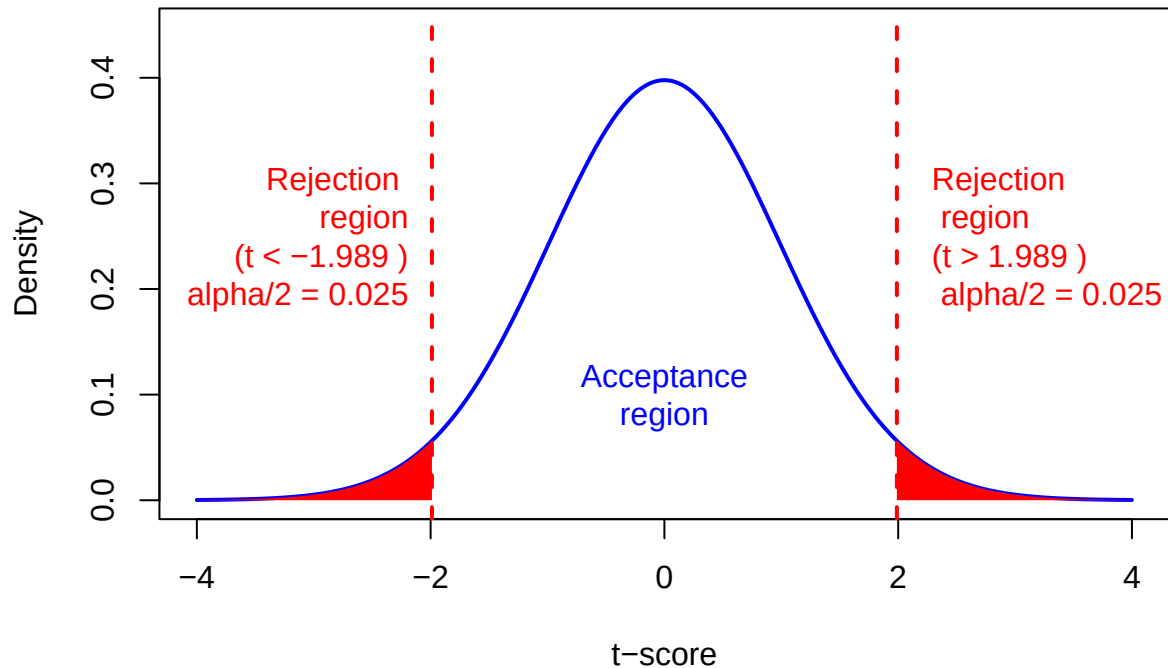
$$TS = \frac{\bar{x}_A - \bar{x}_B - 0}{s_p \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}}$$

$$TS = \frac{193.44 - 184.32 - 0}{31.11 \sqrt{\frac{1}{40} + \frac{1}{45}}} = 1.34$$

Rejection Region

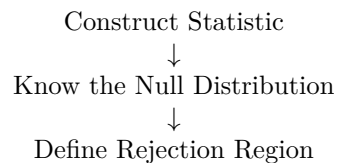
For a two-tailed test we are concerned if the difference is either too large or too small. Thus, we split significance probability equally between both tails. On a t-distribution with $40 + 45 - 2 = 83$ df, the 2.5% upper tail cutoff would be 1.989 and the 2.5% lower tail cutoff would be -1.989.

t-Distribution (df = 83) with Two-Tailed 5% Rejection Regions



Conclusion: Our test statistic is 1.34 which falls in the acceptance region. We cannot conclude that there is a difference between the new inference and the old (control) inference time.

This problem showed same framework, different null distribution:



p-values

Definition: The **p-value** is the probability under a specified statistical (H_0) model that a statistical summary of the data (for example, the sample mean difference between two compared groups) would be equal to or more extreme than its observed value.

To decide whether or not to reject H_0 , we compare the p-value to our significance level.

$\begin{aligned} \text{p-value} < \alpha &\rightarrow \text{reject } H_0 \\ \text{p-value} > \alpha &\rightarrow \text{do not reject } H_0 \end{aligned}$

A p-value is simply another way of describing the rejection region.

Common misunderstandings:

In 2014 George Cobb (Professor Emeritus at Mount Holyoke College) posted some questions regarding p-values to the American Statistical Association (ASA) discussion forum. Because of his questions, it prompted the ASA to write the following position paper.

Wasserstein, R. L., & Lazar, N. A. (2016). The ASA Statement on p-Values: Context, Process, and Purpose. The American Statistician, 70(2), 129–133. <https://doi.org/10.1080/00031305.2016.1154108>

I suggest you read the paper, but here is a summary of the findings:

- P-values can indicate how incompatible the data are with a specified statistical model.
- P-values do not measure the probability that the studied hypothesis is true, or the probability that the data were produced by random chance alone.
- Scientific conclusions and business or policy decisions should not be based only on whether a p-value passes a specific threshold.
- Proper inference requires full reporting and transparency.
- A p-value, or statistical significance, does not measure the size of an effect or the importance of a result.
- By itself, a p-value does not provide a good measure of evidence regarding a model or hypothesis. Data science professionals should recognize that a p-value without context or other evidence provides limited information.

So what do p-values have to do with rejection regions?

Suppose we have an α -level test. If the p-value falls in the rejection region, the p-value will be less than α . Otherwise, it will be greater than α .

Consider the two-sample t-test we just performed. Our test statistic was 1.34 which fell in the acceptance region for $\alpha = 0.05$.

The p-value for that test would be $\mathbb{P}(T > 1.34) + \mathbb{P}(T < -1.34)$, where $T \sim t_{83}$. Because of symmetry, we only need to calculate one of the 2 probabilities and then double it.

```
pt(1.34,df=83,lower.tail=FALSE)*2
```

```
## [1] 0.1839024
```

Our p-value exceeds the $\alpha = 0.05$ threshold, so we wouldn't reject the null hypothesis.

$$\mu = \mu_0 \text{ not in CI reject } H_0 : \mu = \mu_0 \text{ at level } \alpha$$

Connection between CIs and Hypothesis tests

Confidence Intervals provide a **range of plausible values**

Interpretation:

A $(1 - \alpha)$ confidence set $C(X)$ satisfies: $P_\theta(X : \theta \in C(X)) \geq 1 - \alpha$ for all $\theta \in \Theta$.

If we were to repeatedly sample data from P_θ and construct a confidence interval each time using the same procedure, then in the long run at least $100(1 - \alpha)\%$ of those intervals would contain the true parameter θ .

Example (Normal mean):

Let $X_1, X_2, \dots, X_n$ be an iid sample from $N(\mu, \sigma^2)$, $\sigma^2 < \infty$ is known.

100(1 - α)% Confidence Interval:

$$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

This results in a Lower Bound = $\bar{x} - Z_{\alpha/2}$ and Upper Bound = $\bar{x} + Z_{\alpha/2}$

α -level Hypothesis Test.

$$H_0 : \mu = \mu_0, H_1 : \mu \neq \mu_0$$

If we test any value of μ_0 between $\bar{x} - Z_{\alpha/2}$ and $\bar{x} + Z_{\alpha/2}$ the null hypothesis would not be rejected. If μ_0 is outside this range it would be rejected.

Randomized vs. non-randomized tests

Based on study design the independent variable categorizes can be randomized or not. In a clinical drug trial, the independent variable would be whether or not you received a drug or a placebo. This is normally done through randomization. This means that if you are a study participant, you do not know which you are getting. Sometimes even the doctor wouldn't know (i.e. double blind study). Randomization is good, because when performing a hypothesis test we would like to know that if the result is significant, we want to know if that was because of the treatment effect and not some other factor.

Another factor that could influence the outcome is called a **confounding variable**. By randomizing, we randomly place people in groups thereby spreading the confounding variables equally (or almost equally) between the groups so the only difference now between the groups is the treatment effect.

Sometimes though, we are not at liberty to randomize. Suppose you wanted to prove that a company paying women lower salaries than men. You cannot randomly put people in groups (i.e. male/female), so stating that the group assignment affected the outcome is much more difficult.

Summary

Hypothesis testing is a foundational tool in statistical inference that remains highly relevant in modern data science—despite the rise of Bayesian methods, machine learning, and big data. It provides rigorous control of error rates, interpretable decision frameworks, and direct connections to confidence intervals and multiple testing corrections.

Topic	Description	Key Points
Test Statistics $T = T(X)$	We reduce high-dimensional data X to a low-dimensional (often scalar) summary T that captures evidence against the null hypothesis H_0 .	Why reduce data: Summarizes evidence efficiently.- Invariance intuition: Designed to be invariant (or equivariant) under transformations that leave the testing problem unchanged (e.g., location shifts or scaling), eliminating nuisance parameters.
Null Distribution	The sampling distribution of T assuming H_0 is true.	Pivotal quantities: Have known, parameter-free null distributions (e.g., t-statistic), enabling exact inference even with unknowns like σ .
Rejection Regions	Pre-specified sets of T values where we reject H_0 .	Size control: Controls Type I error rate (size α , usually 0.05).- One- vs two-sided: One-sided for directional alternatives; two-sided for non-directional (common in exploratory data science).

Topic	Description	Key Points
Significance Level α	The probability of rejecting H_0 when it is true (false positive rate).	Fixed in advance to control errors.
p-values	The probability, under H_0 , of observing a test statistic at least as extreme as the one obtained.	Correct interpretation: Measures incompatibility with H_0 ; not $\mathbb{P}(H_0 \text{ is true})$. ASA statement (2016): p-values indicate evidence against H_0 , but do not quantify hypothesis probability or effect size.
CI/Test Connection	Duality between confidence intervals (CIs) and tests.	$(1 - \alpha)$ CI contains parameter values not rejected at level α .

Key Takeaways for Data Science PhD Students

- **Error control matters at scale.** In high-throughput settings (genomics, A/B testing, ML feature selection), controlling false positives via hypothesis testing (or adjusted p-values/FDR) is often more reliable than pure predictive performance metrics.
- **p-values are evidence, not truth.** Misinterpretation of p-values as $\mathbb{P}(H_0|\text{data})$ is widespread. Always report effect sizes and confidence intervals alongside p-values.
- **Classical tests are building blocks.** Even when using complex models, you'll frequently fall back on t-tests, z-tests, or chi-squared tests for quick sanity checks, model diagnostics, or reporting in papers.
- **Think duality.** Confidence intervals are often more interpretable than p-values alone. A non-significant p-value does not mean "no effect"; it may just mean insufficient evidence.
- **Be cautious with assumptions.** Normality and independence assumptions underlie exact classical tests. In real data science, robustness checks (non-parametric alternatives, bootstrap) are essential.
- **Why do we always work under H_0 ?** Because our optimization problem constrains Type I risk under the null.